

Figure 4: Cross-disciplinary analogies — earthquakes, neuronal avalanches, non-reciprocal phases Code description for GitHub repository (Mevan Rajakaruna)

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Overview

This Python script generates Figure 4 of the manuscript entitled “Feedback polarity controls causal clustering and cascade propensity in stochastic rupture”. The figure comprises four panels (a–d) that illustrate conceptual analogies between the feedback polarity framework developed in this work and three other physical systems: earthquake physics, neuronal avalanches, and non-reciprocal phase transitions in active matter. Panel d provides a summary schematic synthesising the common theme that feedback polarity—positive or negative—determines whether collective spreading or localised behaviour emerges. The figure is purely schematic and involves no numerical simulations or data fitting.

1 Software and libraries

The script requires Python version 3.9 or higher. Numerical array operations are handled by NumPy (version 1.21 or later). All figure construction, including patches, circles, rectangles, annotations, and file saving, is performed using Matplotlib (version 3.4 or later). Specific patch classes imported include `Circle`, `Rectangle`, and `FancyBboxPatch` from `matplotlib.patches`, as well as `mlines` for basic line elements.

2 Plotting style

Publication-quality visual styling is enforced through customisation of `matplotlib.rcParams`. The font family is set to sans-serif (Arial or Helvetica), with a base font size of 9 points. Axes labels are 10 points, titles 10 points, tick labels 8 points, and legend text 8 points. The figure is rendered at 300 DPI for screen display and saved at 600 DPI. The bounding box is tight with a padding of 0.05 inches. Unlike previous figures, no specific line width customisations are applied beyond the defaults.

3 Figure layout

The figure has dimensions 9.2 inches wide by 6 inches high. A 2×3 grid is created using `fig.add_gridspec` with height ratios `[1,1.2]`, horizontal spacing of 0.18, and vertical spacing of 0.28. Panels a, b, and c occupy the three columns of the first row. Panel d occupies the entire second row.

4 Panel a: Earthquake physics

Panel a illustrates cascading rupture in a fault system under tectonic displacement control. The axis limits are $x \in [0.5, 7.2]$ and $y \in [0, 4]$, with the axis turned off. A black horizontal line of width 3 at $y = 2$ represents the fault. Two tectonic plates are drawn as light gray rectangles with black edges: the left plate spans $x \in [0.8, 3.4]$ and $y \in [1.5, 2.5]$, and the right plate spans $x \in [4.6, 7.2]$ and $y \in [1.5, 2.5]$. Plate motion is indicated by a blue arrow from (3.1, 2.5) to (2.0, 2.5) labeled “Plate motion” in blue, and a red arrow from (5.1, 1.5) to (6.3, 1.5) labeled “Plate motion” in red.

The rupture initiation site is marked by a red circle of radius 0.2 at (3.2, 2), with an annotation t_1 above it. Three concentric orange dashed circles centered at the initiation site represent seismic wave fronts, with radii 0.4, 1.2, 2.0 (i.e., $i \times 0.8 + 0.4$ for $i = 0, 1, 2$), an opacity of 0.65, and linewidth 1.5. A second rupture site is marked by a yellow circle of radius 0.2 at (5.7, 2), annotated $t_2 = t_1 + \tau$ at (5.85, 2.28). The distance between rupture sites L is indicated by a double-headed black arrow between (3.4, 1.5) and (5.5, 1.5), with the label L at (4.45, 1.28). A light yellow box at (4.45, 0.52) contains the text “Cascading if $\tau < L/c_s$ ”. The panel title “Earthquake physics” is centered at (4.0, 3.7) in bold font. The panel label “a” is placed at (0.02, 0.98) in the axes coordinate system.

5 Panel b: Neuronal avalanches

Panel b illustrates neuronal avalanche dynamics with excitatory (positive feedback) and inhibitory (negative feedback) connections. The axis limits are $x \in [1.0, 7.0]$ and $y \in [0, 4]$, with the axis turned off. Five neurons are positioned at coordinates (2, 2), (4, 1.5), (6, 2.5), (3, 3), and (5, 1). Each neuron is drawn as a light blue circle of radius 0.3 with a blue edge of linewidth 1.5, and labeled N1 through N5 in 6-point font.

Excitatory (positive feedback) connections are drawn as green arrows with linewidth 1.5 and opacity 0.7 along the following paths: from N1 (2, 2) to N2 (4, 1.5); from N1 (2, 2) to N4 (3, 3); from N2 (4, 1.5) to N3 (6, 2.5); and from N4 (3, 3) to N5 (5, 1). An inhibitory (negative feedback) connection is drawn as a red arrow of linewidth 2 and opacity 0.7 from N3 (6, 2.5) to N5 (5, 1). The excitatory connections are labeled “Excitatory (positive fb)” in green at (2.05, 2.55), and the inhibitory connection is labeled “Inhibitory (negative fb)” in red at (6.10, 1.5).

An initial firing event is indicated by an orange dashed circle of radius 0.4 around N1 at (2, 2), with the text “Initial spike” below at (2, 1.2). An avalanche region is drawn as a rounded light coral rectangle with opacity 0.18 and a red edge, spanning $x \in [3.55, 6.5]$ and $y \in [1.0, 3.0]$. The annotation inside the avalanche region reads “Avalanche if $\tau < t_{\text{mat}}$ ” at (5.0, 2.05). The panel title “Neuronal avalanches” is centered at (4.0, 3.7) in bold font. The panel label “b” is placed at (0.02, 0.98) in the axes coordinate system.

6 Panel c: Non-reciprocal phase transitions

Panel c illustrates non-reciprocal interactions in active matter systems. The axis limits are $x \in [1.3, 7.9]$ and $y \in [0, 4]$, with the axis turned off. Two bonds are drawn as light blue circles of radius 0.5 at (3, 2) and (5, 2) with blue edges of linewidth 2. The circles are labeled “Bond 1” and “Bond 2” respectively in 6-point font.

A strong (positive feedback) coupling is shown as a red arrow of linewidth 3 from Bond 1 to Bond 2, along the path $(3.5, 2.2) \rightarrow (4.5, 2.2)$, annotated with “Strong coupling (positive feedback)” in red at $(4, 2.72)$. A weak or absent (no feedback) coupling is shown as a blue arrow of linewidth 1 and opacity 0.5 from Bond 2 to Bond 1, along the path $(4.5, 1.8) \rightarrow (3.5, 1.8)$, annotated with “Weak/no coupling (no feedback)” in blue at $(4, 1.13)$. The text “Failed at t_1 ” appears at $(3, 0.90)$ in bold, accompanied by a small red circle of radius 0.1 at $(3, 0.7)$ indicating bond failure. A purple arrow of linewidth 2 from $(6.4, 2)$ to $(7.2, 2)$ indicates the transition to a collective phase, with the label “Collective phase” in purple at $(6.8, 2.3)$. The panel title “Non-reciprocal phases” is centered at $(4.0, 3.7)$ in bold font. The panel label “c” is placed at $(0.02, 0.98)$ in the axes coordinate system.

7 Panel d: Summary schematic

Panel d synthesises the three analogies into a unified framework connecting feedback polarity to universality class. The axis limits are $x \in [0, 10]$ and $y \in [0, 4]$, with the axis turned off. Three columns are centered at $x = 2, 5, 8$ with labels “Earthquake physics”, “Neuronal avalanches”, and “Non-reciprocal phase transitions” respectively. Each column contains a light gray rectangle of opacity 0.2 with a black edge of linewidth 0.8, spanning $x \in [x_0 - 1.5, x_0 + 1.5]$ and $y \in [0.5, 3.5]$.

For the earthquake physics column ($x_0 = 2$), the text “Positive feedback” appears in red at $y = 2.5$, and “Cascading rupture” appears at $y = 1.5$. A red double-headed arrow between $(1.5, 2.2)$ and $(2.5, 2.2)$ indicates positive feedback.

For the neuronal avalanches column ($x_0 = 5$), the text “Excitatory: +” appears in green at $y = 2.7$, and “Inhibitory: -” appears in red at $y = 2.2$. The text “Avalanches/Suppression” appears at $y = 1.5$. A green arrow from $(4.5, 2.9)$ to $(5.5, 2.9)$ indicates excitatory (positive) coupling, and a red arrow from $(5.5, 2.4)$ to $(4.5, 2.4)$ indicates inhibitory (negative) coupling.

For the non-reciprocal phases column ($x_0 = 8$), the text “Asymmetric feedback” appears in purple at $(8.35, 2.5)$, and “Collective phase” appears at $y = 1.5$. A purple arrow from $(7.7, 2.8)$ to $(8.3, 2.2)$ indicates the asymmetric coupling that drives collective behaviour.

A light yellow box at the bottom of panel d, centered at $(5, 0.2)$, contains the text “↓ Feedback polarity determines universality class ↓” in bold 9-point font. The panel label “d” is placed at $(0.02, 0.98)$ in the axes coordinate system.

8 Output files

Despite the panel labels indicating Figure 4, the script saves the figure with filenames containing “fig3”. Specifically, the script saves three files: `fig3_analogies.eps` (Encapsulated PostScript), `fig3_analogies.png` (PNG at 600 DPI), and `fig3_analogies.pdf` (Portable Document Format). All files are saved in the current working directory. The figure is also displayed interactively via `plt.show()`, and the script prints the message “Figure 3 saved as fig3_analogies.eps, .png, .pdf”. Users should note the discrepancy between the intended figure number (4) and the saved filenames (fig3); renaming may be required for manuscript submission.

9 Reproducibility

No random numbers are used; the figure is entirely deterministic. No external data files are required. To regenerate the figure exactly, the script must be executed in a Python environment with the specified library versions.

10 Relation to the manuscript

Panel a corresponds to Figure 4a in the paper, illustrating earthquake physics where tectonic displacement control provides positive feedback and cascading rupture occurs when $\tau < L/c_s$. Panel b corresponds to Figure 4b, illustrating neuronal avalanches where excitatory connections provide positive feedback and inhibitory connections provide negative feedback. Panel c corresponds to Figure 4c, illustrating non-reciprocal phase transitions in active matter where asymmetric feedback can produce collective phases. Panel d corresponds to Figure 4d, synthesising the common organising principle that feedback polarity selects the universality class of collective behaviour.

License and citation

This code is part of the research by M. Rajakaruna and H. Rajakaruna. If you use this code or the generated figure, please cite the associated manuscript as described in the `README.md` file in the repository.